

AI-Generated Image Detection

A Comparative Study of CNN Architectures

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Applied Artificial Intelligence: Computer Vision

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Team Contributions

- Victor Salcedo:
Full EDA, preprocessing verification, report writing
- Marston Ward:
Custom CNN development & training
- Jasper Dolar:
ResNet50 model & secondary model

Dataset & Methodology

Dataset Characteristics

Source: Hugging Face

Total: 152,710 images

Classes: Real (0) vs AI-Generated (1)

Split: 122,168 train / 15,271 val / 15,271 test

Balance: Equal class distribution

Diversity: Portraits, animals, objects, scenes

Preprocessing:

- Resize to 224×224 pixels (RGB)
- ImageNet normalization
- Random flips, rotations, color jitter

Training Configuration

Hardware: A100 GPU

Precision: AMP (Mixed)

Batch Size: 32

Workers: 2

Test Set: 15,271 images

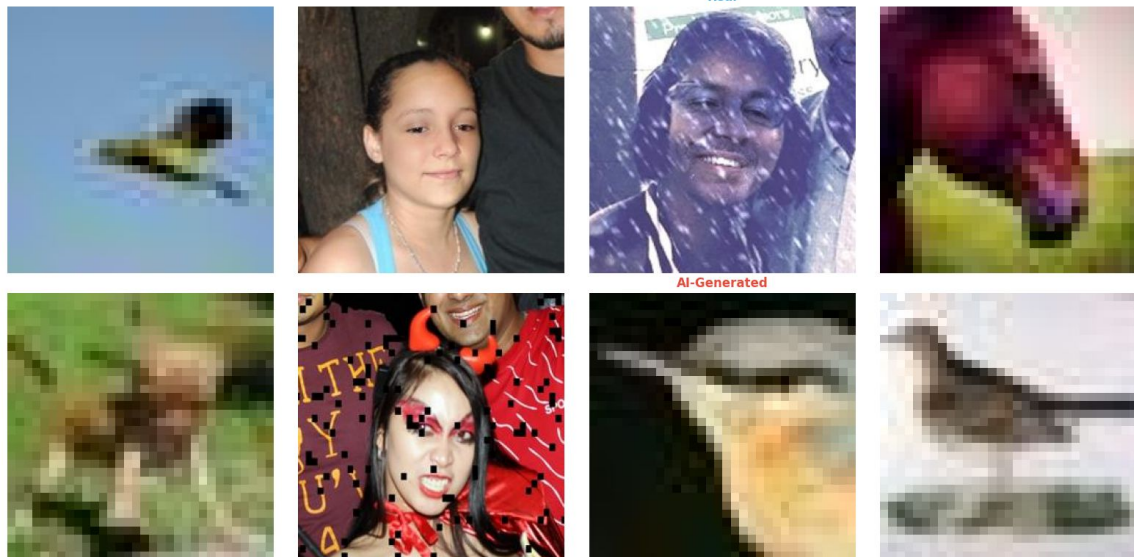
Key Decisions:

- Balanced dataset reduces bias
- ImageNet normalization supports transfer learning
- Conservative augmentation preserves features

Class Distribution



Sample Images: Real vs AI-Generated



DataLoader Verification

- Batches formed correctly
- Labels aligned with images
- Augmentation applied only to training
- Validation/Test sets kept clean
- Normalization reversible for visualization

EDA Summary

- Dataset is balanced and diverse
- Preprocessing preserves essential features
- Dataset large enough for deep learning
- Visual checks confirm labeling accuracy
- Ready for CNN and ResNet model training